Blessed G/Michael Catholic Secondary School
1st Semester Chemistry Final for Grade 11th

January, 2018 E.C

Time Allowed - 1:40

Name Aluhamis Menale NO 33 Section 11C

Physical constants;

Speed of light, $c=3*10^8 m/s$

Planck's constant, $h=6.626*10^{-34} JS$

$1eV = 1.60218 \times 10^{-19} J$

Mass of an electron $/m_e/=9.11*10^{-31} Kg$

Rydberg's constant$/R_H/=2.18*10^{-18} J$

I. Choose the best answer from the given alternatives.

B 1. What is the wave length (in meter) of the first line in the Lyman series?

A. $\frac{36hC}{5RH}$ B. $\frac{4hC}{3RH}$ C. $\frac{56hC}{9RH}$ D. $\frac{4hC}{RH}$

A 2. No two electrons have the same set of all four quantum numbers.

A. True B. False

D 3. How many 3d electrons are present in the ground state of copper atom (z=29)?

A. 4 B. 5 C. 9 D. 10

B 4. How many unpaired electrons are there in the $_{24}Cr^{3+}$ ion?

A. 6 B. 3 C. 1 D. 0

A 5. Antimony (Sb) is a group "VA" element. What will be the molecular geometry and number of lone pair Electrons, respectively that exists in the ion $[SbCl_4]^-$?w 5+1+28=34 32 [illegible]

A. See saw, 1 B. square planar, 2 C. see saw, 2 D. linear, 3

B 6. What is the fundamental principle of Valence Bond Theory (VBT) for covalent bond formation?

A. Atomic orbitals merge to form new molecular orbitals

B. Bonds form through the overlap of partially filled atomic orbitals.

C. Electrons are delocalized across the entire molecule

D. Hybridization is essential for all stable bonds.

B 7. What is the hybridization and geometry for IF_4^-? 7+1+28=36 32 2lp 4bp

A. sp^3d^2, octahedral C. sp^2, trigonal planar

B. sp^3d^2, square planar D. sp^3d, trigonal bipyramidal

D 8. In VBT, which type of atomic orbital overlap results in the formation of a sigma (σ) bond?

A. Side-by-side overlap C .Overlap involving only 's' orbitals

B. No overlap is required for sigma bonds D. End-to-end (head-on) overlap

sp
4+12=16 4+4=8
sp³

C 9. The type of hybrid orbitals of nitrogen in NO_2^+, NO_3^-, NH_4^+ respectively are expected to be ;

A. Sp^2, Sp^3 and Sp
B. Sp^2, Sp & Sp^3
sp C. Sp, Sp^2 & Sp^3
D. Sp, Sp^3, Sp^2

D 10. The number of orbitals with n = 5, $m\ell$ = 2 is ;

A. 6
B. 1
C. 2
D. 3
E. 8

C 11. The energies ϵ_1 and ϵ_2 of two radiations are 25ev & 50ev respectively the relation between their wave length , ℓ_1 & ℓ_2 will be ;

A. $\ell_1 = \frac{\ell_2}{2}$
B. $\ell_1 = \ell_2$
C. $\ell_1 = 2\ell_2$
D. $\ell_1 = 4\ell_2$

A 12. The 1st, 2nd, 3rd, 4th and 5th ionization potential values of elements are 6.11, 11.87, 51.21,67, and 84.37ev respectively. The element is

A. $_{20}$Ca
B. $_{19}$K
C. $_{13}$Al
D. $_6$C

C 13. Which intermolecular force is present in all molecules, including nonpolar ones like Argon (Ar)?

A. Hydrogen Bonding
B. Dipole-Dipole Interactions
C. London Dispersion Forces (LDF)
D. Ion-Dipole Forces

B 14. Why are metals malleable and ductile (can be hammered into sheets and drawn into wires)?

A. Metallic bonds are directional and break easily when stressed.
B. The sea of electrons allows layers of metal ions to slide past each other without breaking the bond.
C. Metal atoms are small and easily rearranged.
D. Metals have high melting points, making them malleable.

C 15. Which factor generally increases the strength of a metallic bond?

A. A larger atomic radius
B. Fewer delocalized valence electrons
C. A higher charge on the metal ions and more delocalized electrons
D. Weaker attraction between ions and the electron sea

D 16. Which of the following statements is not true regarding the electron sea model for the electronic structure of metals?

A. Electrons are static and individual electron is confined to a particular metal ion
B. Electrons are uniformly distributed throughout the structure A
C. The nucleus of each atom is surrounded by the outer electrons of all the atoms.
D. The electrons are confined to the metal by electrostatic attraction of the cations.

C 17. Each P- orbital can accommodate a maximum of ------------electron.

A. 6
B. 3
C. 2
D. 1

B 18. Region of space around the nucleus of an atom that can be occupied by one or two electrons are called ____

A. Shells
B. orbitals
C. energy levels
D. orbits

A 19. Which of the following molecules would you expect to be non- planar?

A. CH_4 B. BCl_3 C. XeF_4 D. HCHO

C 20. Which of the following explains why at room temperature, I_2 is a solid, Br_2 is liquid and Cl_2 is a gas?

A. Hydrogen bonding C. London dispersion force

B. Ionic bonding D. Dipole- dipole force

A 21. Which of the following is not a property of an ionic compound?

A. malleable B. Brittle C. Hard C. high meting point

D 22. Which hybridization change. If any, occurs at the underlined atom in the following reaction?

$\underline{S}O_2+O_2 \longrightarrow \underline{S}O_3$ O-S-O 6+12=24

A. $sp^2 - sp^3$ B. $sp^3 \rightarrow sp^3d$ C. $sp \rightarrow sp^2$ D. No hybridization change observed

C 23. Which of the following is not the correct geometric configuration for the given molecule?

A. BF_3, trigonal planar ✓ B. SF_6, octahedral ✓ C. NF_3, trigonal planar ✓ D. CH_4, tetrahedral

B 24. The maximum kinetic energy of photoelectrons emitted from a metal is $1.03*10^{-19}$J when light that has a 656r wavelength shines on the surface .what is the threshold frequency for this metal?

A $4.57*10^{14}s^{-1}$ B. $3.02*10^{14}s^{-1}$ C. $3.02*10^{-14}s^{-1}$ D. $4.57*10^{-14}s^{-1}$

A 25. The threshold wave length of photoelectric effect of a metal is 6500 angstrom. The work function of the metal is approximately.

A. 2ev B. 1ev C. 0.1ev D. 3ev

B 26. Two of the species have the same shape $NI_3, I_3^-, SO_3^{2-}, NO_3^-$ these are:-

A. NI_3, I_3^- B. NI_3, SO_3^{2-} C. SO_3^{2-}, NO_3^- D. I_3^-, NO_3^-

A 27. Consider the two structures

I. F_2Cl^+ 2bp 2lp

II. F_2Cl^- select the correct statement 2bp 3lp

A. I is bent, II is linear C. Both are linear

B. I is linear, II is bent D. Both are bent

D 28. Which molecule listed below has two sigma and two pi- bonds?

A. N_2 B. C_2H_4 C. N_2F_2 D. HCN

B 29. Which of the following is not true about the Bohr's atomic postulate?

A. The energy of an electron remains the same as long as it stays in the same orbit.

B. When electrons fall from a higher state they absorb fixed amount of energy. release

C. Electrons release or absorb energy when they move between orbits.

D. Electrons revolve around the nucleus in circular orbits.

D 30. Once formed, how are coordinate covalent bonds differ from other covalent bonds?

A. They are stromer C. They are weaker

B. They are more ionic in character D. there is no difference

C 31. From CO_2, H_2O, $BeCl_2$ and N_2O_5 which have the same molecular geometry?

A. CO_2, H_2O and N_2O C. CO_2, $BeCl_2$ and N_2O

B. CO_2 and $BeCl_2$ only D. H_2O and N_2O only

B 32. Which reaction involves a change in electron pair geometry for the underlined atom?

A. $\underline{N}H_3 + H^+ \rightarrow \underline{N}H_4^+$ B. $\underline{B}F_3 + F^- \rightarrow \underline{B}F_4^-$ C. $2\underline{S}O_2 + O_2 \rightarrow 2\underline{S}O_3$ D. $H_2\underline{O} + H^+ \rightarrow H_3\underline{O}^+$

C 33. What is the wave length associated with an electron of mass, m= $9.11 * 10^{-28}$ gram, travelling at 40% of the velocity of light?

A. $6.06 * 10^{-15}$m B. $2.42 * 10^{-15}$m C. $6.06 * 10^{-12}$m D. $2.42 * 10^{-11}$m

D 34. Which of the following substances exhibits hydrogen bonding?

A. CH_4 B. HCl C. H_2S D. NH_3

C 35. Which of the following transition will emit maximum energy in the H- atom?

A. n=4 → n=2 B. n=4 → n=3 C. n=2 → n=1 D. n=3 → n=2

D 36. Which of the following is a property of covalent compounds?

A. *Has high melting* and boiling points C. Is crystalline solids at room temperature

B. *Is non- volatile* D. Is generally insoluble in water

37. Formic acid, which is released by ants, has a molecular formula of HCOOH. What are the possible hybridization that exist in the molecule?

A. SP^2 and SP^3 B. SP and SP^3 C. SP, SP^2 and SP^3 *D. SP and SP^2*

B 38. Which model proposed that electrons orbit the nucleus in fixed energy levels?

A. Plum Pudding Model B. Bohr Model C. Rutherford Model *D. Quantum Model*

B 39. How many sigma and pi bonds are present in the following molecule?

$H_2C\text{-}CH = CH\text{-}CH_3$

A. 8σ bonds and 1π bond C. 8σ bonds and 2π bond

B. 10σ bonds and 2π bond D. 11σ bonds and 1π bond

D 40. Which of the following molecule does not have a trigonal bipyramidal electron pair geometry?

A. SF_4 B. ClF_3 C. XeF_2 D. BrF_5

T-shaped linear

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Jan, 2018 E.C

TIME ALLOWED: 1:40'

Name [illegible] Menale grade 11 Section C no 33

1. B
2. A
3. D
4. B
5. A
6. B
7. B
8. D
9. C
10. D
11. C
12. A
13. C
14. B
15. C
16. ~~D~~A
17. C
18. B
19. A
20. C
21. A
22. D
23. C
24. B
25. A
26. B
27. A
28. D
29. B
30. D
31. C
32. B
33. C
34. D
35. C
36. D
37. A
38. B
39. D
40. D

39/40

Excellent